UNIVERSITY OF MIAMI
Curriculum Vitae

1. *Date:* Nov 9th, 2026

Personal

2. Name: Kyle Edward Grealis

3. Email Address: kylegrealis@miami.edu

4. Current Academic Rank: Lead Data Analyst / Biostatistician

5. Primary Department: Dept. of Public Health Sciences, Division of Biostatistics and Bioinformatics, University of Miami Miller School of Medicine

6. Secondary or Joint Appointments: N/A

Higher Education

7. Institutional:

2023: University of Miami Miller School of Medicine, Miami, FL; M.S., Biostatistics (Summa Cum Laude)
2020: Purdue University Global, West Lafayette, IN; B.S., Health Sciences (Summa Cum Laude)
2002: Rutgers University (f.k.a. UMDNJ), Stratford, NJ; A.A.S., Respiratory Therapy

8. Non-Institutional:

2025: CITI Program; Good Clinical Practice, Biomedical Researchers, Conflict of Interest
2022: DataCamp; Data Scientist with R Track

9. Certification, licensure:

2022: SAS Certified Statistical Business Analyst Using SAS 9: Regression and Modeling
2022: American Heart Association; Advanced Cardiac Life Support (ACLS)
2022: American Heart Association; Basic Life Support (BLS)
2021: Florida Department of Health; Registered Respiratory Therapist
2021: SAS 9.4 Base Programming Certification
2021: SAS Certified Associate: Programming Fundamentals
2021: American Heart Association; Pediatric Advanced Life Support (PALS)
2019: National Board for Respiratory Care (NBRC); Registered Respiratory Therapist
2012: American Heart Association; Basic Life Support Instructor
2004–2019: Florida Department of Health; Certified Respiratory Therapy Technician
2003: New Jersey State Board of Respiratory Care; Certified Respiratory Therapy Technician
2003: National Board for Respiratory Care (NBRC); Certified Respiratory Therapy Technician

Experience

10. Academic:

2023–present: University of Miami, Department of Public Health Sciences, Miami, FL; Lead Research Analyst
Responsibilities: Implementation of machine learning models on population health studies; development of data pipelines from REDCap to end users; data visualization and Shiny web application development; coauthoring R packages for data science workflows.

2022–2023: University of Miami, Miami, FL; Bioinformatics Analyst
Responsibilities: Data cleaning, exploratory analysis, and statistical modeling for international pulmonary research studies; contributed to statistical analysis plans and academic manuscripts; incorporated SAS/STAT, SAS/GRAPH, SAS/MACRO, and R throughout the analytic process; performed pulmonary function testing per ATS protocols.

11. Non-Academic:

2019–2022: University of Miami, Miami, FL; Senior Respiratory Therapist
Responsibilities: Pulmonary function testing; assisted with rigid and fiberoptic bronchoscopy with endobronchial ultrasound; developed in-patient and out-patient plans of care; coordinated home rehabilitation therapies.

2004–2019: Physicians Regional Medical Center, Naples, FL; Registered Respiratory Therapist
Responsibilities: Clinical Coordinator; critical care evaluation and planning; new employee and student preceptorship; BLS instructor.

2003–2004: Kennedy Health Systems, Stratford, NJ; Certified Respiratory Therapist

1998–2003: Kennedy Health Systems, Stratford, NJ; Pharmacy Technician

12. Military:

2017–2020: United States Army Reserve, 365th Engineer Company (SAPPER), Cape Coral, FL; Equal Opportunity Leader

2016–2020: United States Army Reserve, 365th Engineer Company (SAPPER), Cape Coral, FL; Squad Leader

Publications

13. Books and Monographs Published: N/A

14. Juried or Refereed Journal Articles or Exhibitions:

1. Balise RR, **Grealis K**, Brandt L, Luo SX, Odom G, Jean-Baptiste G, Feaster DJ. (2026). Applying machine learning in predicting medication treatment outcomes for opioid use disorder. *Journal of Substance Use and Addiction Treatment*, 182: 209847 (1-8). https://doi.org/10.1016/j.josat.2025.209847
2. Harkness A, Morales V, **Grealis K**, Reyes N, Feaster DJ, Safren S, Turner D, Balise RR. (2025). Implementation determinants of PrEP and behavioral health treatment referral among HIV test counselors. *AIDS & Behavior*, 29(5): 1492–1506. https://doi.org/10.1007/s10461-025-04620-2

3. Poudel L, Baral A, Abdshah A, **Grealis K**, Aka A, Paudel S, Vidot DC. (2025). Sleep disturbances among young adult dual users of cigarettes and e-cigarettes: analysis of the 2020 National Health Interview Survey. *PLOS ONE* 20(3), e0320748. https://doi.org/10.1371/journal.pone.0320748
4. Goddard-Eckrich D, **Grealis K**, El-Bassel N, Lounsbury DW, Dsouza N, Bhuiyan J, Cervantes M, Angerame A, Feaster DJ, Kim E, Huang TTK, Sabounchi NS, Gilbert L, Levin FR, Edwards K, Gatanaga OS, McCrimmon T, David JL, Hunt T, Nunes EV, Wu E, Gutnick D, Rodriguez S, Gruss DE, Rodgers E, Campbell ANC, Xu J, Balise R. (2025). Development of a brief stigma and perceptions questionnaire for pharmacists: an exploratory factor analysis approach in New York state counties enrolled in the healing communities study. *Journal of Substance Use and Addiction Treatment*, 169: 209566. https://doi.org/10.1016/j.josat.2024.209566
5. Siddharthan T, **Grealis K**, Robertson NM, Lu M, Liu S, Pollard SL, Hossen S, Jackson P, Rykiel NA, Wosu AC, Flores-Flores O, Quaderi SA, Alupo P, Kirenga B, Ricciardi F, Barber JA, Chandyo RK, Sharma AK, Kumar Das S, Shrestha L, Miranda JJ, Checkley W, Hurst JR et al. (2024). Assessing the prevalence and impact of preserved ratio impaired spirometry in low-income and middle-income countries: a post-hoc cross-sectional analysis. *The Lancet Global Health* 12(9), p.e1498-e1505. https://doi.org/10.1016/S2214-109X(24)00233-X
6. Alexis A, Punjabi NM, **Grealis K**, Wanner A. (2024). A comparative study of bronchodilator response: utilizing pre-bronchodilator versus predicted normal values. *BMC Pulmonary Medicine* 24(1), p.54. https://doi.org/10.1186/s12890-024-02859-4
7. Raju S, **Grealis K**, Gonzalez E, Nakrani K, Otvos T, Kirkness J, Fouras A, Wise R, Punjabi NM, Siddharthan T. (2023). Ventilation heterogeneity from x-ray velocimetry predicts COPD severity. *European Respiratory Journal* 62 (suppl 67), OA4953. https://doi.org/10.1183/13993003.congress-2023.OA4953
8. Siddharthan T, **Grealis K**, Kirkness JP, Otvos T, Stefanovski D, Tombelson A, Dalzell M, Gonzalez E, Nakrani K, Wegner D, Lester MG, Richmond B, Fouras A, Punjabi NM. (2023). Quantifying ventilation by x-ray velocimetry in healthy adults. *Respiratory Research* 24(1), p.215. https://doi.org/10.1186/s12931-023-02517-z

15. Other works, publications and abstracts:

NOTE: All abstracts listed below are for refereed conference presentations.

1. **Grealis K**. (2026). sumExtras: Streamlining Table Workflows in Clinical Research with R. *R/Medicine 2026*, R Consortium. Lightning Talk, May 8, 2026. https://rconsortium.github.io/RMedicine_website/Abstracts.html#sumextras-streamlining-table-workflows-in-clinical-research-
2. **Grealis K**, Balise R. (2025). Mix, Pour, Share: The rUM Cocktail for Biomedical Project Packaging. *R/Medicine 2025*, R Consortium. Regular Talk, June 12, 2025. https://rconsortium.github.io/RMedicine_2025/abstracts.html#rum-cocktail
3. Goddard-Eckrich D, **Grealis K**, El-Bassel N, Lounsbury DW, Dsouza N, Bhuiyan J, Angerame A, Feaster DJ, Cervantes M, Huang TTK, Sabounchi NS, Gilbert L, Medley B, McCrimmon T, David JL, Hunt T, Wu E, Gutnick D, Rodriguez S, Gruss DE, Rodgers E, Nunes EV, Levin FR, Edwards K, Balise R. (2025). Development of the Brief Stigma and Perceptions Questionnaire for Pharmacists: A Confirmatory Factor Analysis

Approach in New York State Counties Enrolled in the Healing Communities Study. *Drug and Alcohol Dependence*, 267: 112166. https://doi.org/10.1016/j.drugalcdep.2024.112166

4. Alexis A, **Grealis K**, Punjabi NM, Wanner A. (2023). Normalizing spirometric bronchodilator responses. *American Thoracic Society*. B30. Breathe (In The Air): Studies in Pulmonary Function, A3002-A3002. https://doi.org/10.1164/ajrccm-conference.2023.207.1_meetingabstracts.a3002
5. Moore DM, **Grealis K**, Gonzalez E, Siddharthan T, Punjabi NM, Abreu AR. (2023). The association between sleep hygiene and cessation of cigarette smoking. *American Thoracic Society*. A36. Sleep-related Health Services Research, A1433-A1433. https://doi.org/10.1164/ajrccm-conference.2023.207.1_meetingabstracts.a1433
6. Tombelson A, **Grealis K**, Otvos T, Kirkness JP, Fouras A, Punjabi NM, Siddharthan T. (2023). Validation of x-ray velocimetry with pneumotachograph to assess respiratory physiology. *American Thoracic Society*. A17. Tell Me What You See: New Studies in Lung Imaging, A1045-A1045. https://doi.org/10.1164/ajrccm-conference.2023.207.1_meetingabstracts.a1045
7. Karmali D, Dalzell M, Gonzalez E, **Grealis K**, Castillo E, Kirkness JP, Otvos T, Fouras A, Punjabi NM, Siddharthan T. (2023). Assessment of regional lung ventilation using computer tomography (CT) lung ventilation analysis in chronic obstructive pulmonary disease (COPD). *American Thoracic Society*. A108. New Insights into Chronic Obstructive Pulmonary Disease, A2699-A2699. https://doi.org/10.1164/ajrccm-conference.2023.207.1_meetingabstracts.a2699
8. **Grealis K**, Siddharthan T, Pollard S, Hossain D, Alupo P, Chandyo R, Cardenas M, Flores-Flores O, Kirenga B, Miranda J, Sharma A, Shrestha L, Checkley W, Hurst J. (2023). Prevalence and burden of PRISm in population cohorts from three LMIC settings. *American Thoracic Society*. C17. In This Together: Confronting Environmental Health Challenges Around the World, A4510-A4510. https://doi.org/10.1164/ajrccm-conference.2023.207.1_meetingabstracts.a4510
9. Siddharthan T, Dalzell M, **Grealis K**, Gonzalez E, Wanner A. (2023). Accuracy and reproducibility of plethysmographic thoracic gas volume measurements in healthy subjects. *American Thoracic Society*. A41. Emerging Clinical and Translational Research in COPD and Alpha-1-Antitrypsin, A1555-A1555. https://doi.org/10.1164/ajrccm-conference.2023.207.1_meetingabstracts.a1555
10. Rovinsky R, **Grealis K**, Kumar N, Velasquez Duran M, Patel S, Ferriera T, Gershengorn HB, Mallow C. (2023). Hemodynamic variations are associated with increased likelihood of discrepancy between pulse oximeter and directly measured oxygen saturation. *American Thoracic Society*. D16. Advancing the Science of ARDS and Acute Respiratory Failure, A6230-A6230. https://doi.org/10.1164/ajrccm-conference.2023.207.1_meetingabstracts.a6230

Published Software Packages:

1. 2026 - ongoing: *nhanesdata* software package for the R language (Harmonized Access to NHANES Survey Data; doi: 10.32614/CRAN.package.nhanesdata) https://cran.r-project.org/package=nhanesdata
2. 2026 - ongoing: *Open.Visualization.Academy* software package for the R language (Content to Support Classes Taught Through the Open Visualization Academy; doi: 10.32614/CRAN.package.Open.Visualization.Academy)

https://cran.r-project.org/package=Open.Visualization.Academy

3. 2025 - ongoing: *sumExtras* software package for the R language (Convenience functions to produce publication quality tables in R; doi: 10.32614/CRAN.package.sumExtras) https://cran.r-project.org/package=sumExtras
4. 2025 - ongoing: *froggeR* software package for the R language (Enhancing reproducibility in Quarto by providing structured templates and methods of reusing metadata across projects; doi: 10.32614/CRAN.package.froggeR) https://cran.r-project.org/package=froggeR
5. 2024 - ongoing: *nascaR.data* software package for the R language (R data package with functions for exploring curated NASCAR datasets of race results and driver performance; doi: 10.32614/CRAN.package.nascaR.data) https://cran.r-project.org/package=nascaR.data
6. 2023 - ongoing: *public.ctn0094data* software package for the R language (De-Identified Data from CTN-0094; doi: 10.32614/CRAN.package.public.ctn0094data) https://cran.r-project.org/package=public.ctn0094data
7. 2022 - ongoing: *rUM* software package for the R language (R templates from the University of Miami; doi: 10.32614/CRAN.package.rUM) https://cran.r-project.org/package=rUM
8. 2020 - ongoing: *tidyREDCap* software package for the R language (Helper Functions for Working with 'REDCap' Data; doi: 10.32614/CRAN.package.tidyREDCap) https://cran.r-project.org/package=tidyREDCap

Published Web Applications:

1. 2024: *EpiMatch*: An automated case-control matching tool for epidemiological studies useful in bias reduction and time management https://kylegrealis.shinyapps.io/case-control-matching/
2. 2024: *PowerViz*: A statistical power calculator and visualization tool written in R Shiny https://kylegrealis.shinyapps.io/powerViz/

16. Other works accepted for publication: N/A

Professional

17. Funded Research Performed:

Pulse Oximetry Bias in Hospitalized Patients
Principal Investigator: Christopher M. Mallow, MD
Protocol / IRB: IRB20220093
Period: 10/2022–present
Role: Data analysis and statistics

ENDS: E-Cigarette and Vascular Diseases
Principal Investigator: Adam Wanner, MD
Protocol / IRB: IRB20211060
Period: 08/2022–present
Role: Data collection/management; data analysis and statistics; behavioral assessment

Airway Blood Flow as an Expression of Airway Inflammation in Cystic Fibrosis
Principal Investigator: Maria Tupayachi Ortiz, MD

Protocol / IRB: IRB20220600
Period: 06/2022–present
Role: Lab testing

Early Detection of Vaping-related Vascular Disease
Principal Investigator: Michael Campos, MD
Protocol / IRB: IRB20211060
Period: 02/2022–present
Role: Data collection/management; data analysis and statistics; behavioral assessment

Effect of Abdominal Gas on Body Plethysmography Measurement of Thoracic Gas Volume
Principal Investigator: Trishul Siddharthan, MD
Protocol / IRB: IRB20210336
Period: 12/2021–07/2022
Role: Data collection

Follow-up Study on Adverse Airway Effects of Inhaled Nicotine from Tobacco and E-cigarettes
Principal Investigator: Michael Campos, MD
Protocol / IRB: IRB20210416
Period: 08/2020–11/2020
Role: Study coordinator

18. *Editorial responsibilities:* N/A

19. *Professional and Honorary Organizations:*

1. 2022: Delta Omega, Beta Sigma Chapter; Honor Society for Graduate Studies in Public Health
2. 2020: International Society of Sports Nutrition, Student Member
3. 2020: National Honor Society
4. 2020: SALUTE Veterans Honor Society (Purdue University Global Chapter)

20. *Honors and Awards:*

1. 2023: University of Miami; Summa Cum Laude
2. 2023: University of Miami; Graduate + Postdoctoral Research Symposium, Poster Winner (Health & Life Sciences)
3. 2020: Purdue University Global; Summa Cum Laude
4. 2020: Purdue University Global; Chancellor's List (four consecutive semesters)
5. 2019: Physicians Regional Medical Center; Employee of the Month (April)
6. 2019: Department of the Army; Army Achievement Medal: distinguished service with 101st Airborne Division
7. 2016: Department of the Army; Leadership Award, Basic Leadership Course
8. 2016: Department of the Army; Distinguished Honor Graduate
9. 2009: Physicians Regional Medical Center; Employee of the Quarter (October)
10. 2002: Rutgers University (f.k.a. UMDNJ); Clinical Excellence Award

21. *Post-Doctoral Fellowships:* N/A

22. Other Professional Activities:

Refereed International Presentations

1. sumExtras: Streamlining Table Workflows in Clinical Research with R. **Grealis K**. *R/Medicine 2026*, R Consortium, online conference. Lightning Talk, May 8, 2026.
2. Mix, Pour, Share: The rUM Cocktail for Biomedical Project Packaging. **Grealis K**, Balise R. *R/Medicine 2025*, R Consortium, online conference. Regular Talk, June 12, 2025.
3. Development of a Brief Stigma and Perceptions Questionnaire for Pharmacists: A Confirmatory Factor Analysis Approach in New York State Counties Enrolled in the Healing Communities Study. Goddard-Eckrich D, **Grealis K**, El-Bassel K, Lounsbury D, Dsouza N, Bhuiyan J, Angerame A, Feaster D, Cervantes M, Huang T, Sabounchi N, Gilbert L, Medley B, McCrimmon T, David J, Hunt T, Wu E, Gutnick D, Rodriguez D, Gruss D, Rodgers E, Nunes E, Levin F, Edwards K, Balise R. *College on Problems of Drug Dependence (CPDD), Annual Scientific Meeting*, Montreal, Canada, June 17, 2024.
4. Validation of x-ray velocimetry with pneumotachograph to assess respiratory physiology. Tombelson A, **Grealis K**, Otvos T, Kirkness JP, Fouras A, Punjabi NM, Siddharthan T. *Young Scientists' Forum*. Oral presentation, May 2023.

Refereed Local Presentations

1. Customizable SAS macro program for case-control matching. **Grealis K**. *Graduate + Postdoctoral Research Symposium*, University of Miami, Miami, FL, March 23, 2023.

Teaching

23. Teaching Awards: N/A

24. Teaching Specialization: N/A

25. Thesis and Dissertation Advising/Post-doctoral Student Supervision: N/A

Service

26. University Committee and Administrative Responsibilities: N/A

27. Community Activities:

Unfunded Software Related Projects:

1. 2020 – Present: Open-Source R & Quarto Software Development (nhanesdata, Open.Visualization.Academy, sumExtras, froggeR, nascaR.data, public.ctn0094data, rUM, tidyREDCap)
   https://github.com/kyleGrealis
   ORCID: 0000-0002-9223-8854